An Approach to Optimizing Kanban Board Workflow and Shortening the Project Management Plan

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***Abstract*—Kanban is a well-known agile method first implemented in Toyota's manufacturing process in the 1950s, later spreading across various industries and academic fields. It focuses on process and project improvement in different fields and emphasizes a team's ability to improve the efficiency of day-to-day activities by implementing the core practices (e.g., visualizing workflow, limiting work-in-progress, managing flow). The efficiency of day-to-day activities is crucial in every process, yet, recent times have called for a new out-of-the-box approach to address the challenges companies face. While the literature review presents efficiency improvements through the use of the core practices, one of the biggest problems while implementing Kanban is shown to be the setting of effective work-in-progress limits. This article addresses this issue and proves that the workflow does not exclusively depend on the work-in-progress limits—as the core practices suggest—but on determining the optimal relationship (solution) among the replenishment value, resource capacity, and work-in-progress limits. Such an approach ensures that the workflow pace on the Kanban board is sustainable by reducing the number of work items in queues as well as people's idleness. The proposed empirical approach runs a series of simulation scenarios aimed at helping to identify the optimal relationship among the three mentioned elements for generating a sustainable workflow pace and minimizing work/people's idleness.**

***Index Terms*—Agile project management, Kanban method, replenishment, resource capacity, work-in-progress (WIP) limits.**

I. Introduction

The COVID-19 pandemic has not only significantly altered the ways companies perform, but also shown that the implementing of certain agile project management practices might be essential for sustaining productivity [1]. The capabilities of teams are, thus, correlated with the efficiency with which they perform their daily activities. The pandemic and other major shocks that have disrupted the predefined and established working modes have been forcing companies to engage in divergent thinking.

This applies to the development of new approaches for application and uses in novel ways with the goal of greater team efficiency. Project management has been a well-researched topic over the decades, with an extensive body of knowledge now being available (e.g., over six million search results in Google Scholar). The use of its methods in the information technology (IT) project management field, in particular, shows that agile principles have been relied on heavily to bring projects to successful completion.

Agile project management consists of various lightweight agile methods that implement agile principles, such as Kanban and Scrum, and is an approach that develops and delivers value-added product increments in iterations. In each iteration, all incremental development stages are executed. Especially, the use of its methods in the IT project management field shows a heavy reliance on agile principles (like a collaborative effort to self-organized cross-functional teams, evolutionary development, early delivery, flexible response to change, and continuous improvement) to bring projects to successful completion.

Specifically, Kanban is an agile improvement method used in many fields, such as in Toyota's manufacturing system, process management, supply chain management, and IT project management. The method makes IT project management works efficiently by implementing several workflow continuous improvements [2]. The Kanban team ensures such improvements by making small changes and implementing gradual improvements in the process, which is also known as Kaizen.

The purpose of this article is to present an approach that may help Kanban teams identify a solution, in order to create a sustainable workflow pace with/without minimal queues on the Kanban board. This can be accomplished by entering a suitable number of work items on the board, which fit the available resource capacity. The approach being introduced optimizes the workflow by establishing a harmonized relationship solution among the three parameters of Kanban—replenishment value, work-in-progress (WIP) limits, and resource capacity.

Our approach uses a simulation as a technique to develop a simulation model that imitates the pulling system, namely, a typical principle in the Kanban method. Running the simulation as an empirical research method is a key step in the approach to establishing the above-mentioned optimal relationship solution. This article's focus is on developing an approach for optimizing workflow on the Kanban board and is not limited to IT projects. The approach may be applied wherever the Kanban method is used in other fields, such as business, healthcare, and manufacturing process management.

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This research article makes the following three contributions to the Kanban method. First, it introduces a new approach to creating efficient flow management on the Kanban board. Second, it proves that to generate a sustainable flow the most effective relationship among replenishment, WIP limits, and resource capacity must be established. Third, the implications held by this article are not only theory-based (by triggering new research on the impact of the Kanban WIP limits principle) but also may be of use to Kanban practitioners (by creating an effective Kanban board and hence changing the management's behavior with respect to the use of agile principles and methods). Similar approaches could be developed to deal with flow efficiency in other agile methods, such as improving the workflow on the Scrum board.

This article has six sections. Section II presents a literature review concentrated on the Kanban method, whereas Section III provides a more detailed description of the method by focusing on its practical implications. Section IV introduces the novel simulation approach, implements it, and shows the results for board simulations. In Section V, a short discussion is presented with regard to the research. Finally, Section VI summarizes the research contribution, sets out conclusions, and presents a few important rules that may assist practitioners.

II. Literature Review

Kanban is made up of two Japanese words—*kan* and *ban*—meaning "a signal card″ and originates from Toyota, which invented it as a scheduling system for just-in-time (JIT) manufacturing in its production system in the 1950s [3]–[6]. Since then, Toyota has spent time perfecting the Kanban method, moving closer to the vision of "one-piece continuous flow" than any other company [3] by emphasizing the reduced timeline between the moment of receiving the customer's order and the moment of collecting the payment for that order by removing nonvalue-added wastes and focusing on minimizing overproduction [7]. Kanban consists of a flow-control mechanism for pull-driven JIT production in which the upstream processing activities are triggered by the downstream process demand signals [8], [9].

A strong practitioner-driven movement supporting the idea of using Kanban in software engineering has emerged in recent decades [10]–[14], having originated in software development in 2004 when Anderson [15] was assisting a small IT team at Microsoft that was operating poorly. Anderson's initiative sparked a growing interest in the Kanban approach, its implementation, and especially the setup within the IT industry (ibid). Al-Baik and Miller [4] conducted a systematic review to analyze the available literature, selecting 37 primary studies from more than 3000 unique studies. They defined the Kanban approach as a set of concepts, principles, practices, techniques, and tools for managing the product development process with an emphasis on the continual delivery of value to customers while promoting ongoing learning and continuous improvements (ibid). In addition, Sjøberg *et al.* [16] suggested using Kanban while dealing with the challenges of effort estimation and interruptions caused by ad hoc bug fixing, support, and maintenance tasks, whereas Majchrzak and Stilger [17] claimed that it supports easy optimization of activities within software projects. Furthermore, in connection with human resources, Kanban is viewed as a tool for the gradual self-improvement of each team (ibid) as new work is pulled into the process only on the completion of the current work and when the resource becomes available [18]. Kanban is seeing ever greater use in software engineering as opposed to other implementations, such as in manufacturing settings [4].

Kanban is seeing ever greater use in software engineering because software companies seek improved project visibility, software quality, team motivation, communication, and collaboration [2] along with customer satisfaction, shorter lead times, and sustainable delivery. A successful Kanban team must ensure continuous improvements (also known as kaizen) by performing small changes and implementing gradual improvements in the identified process since Kanban's objective is to accurately state what work needs to be done and when [6], by prioritizing tasks, defining workflow and lead time to delivery [15]. Hence, the Kanban process stresses the most important tasks that need the most attention to reduce the risk of not being completed, and also to increase the flexibility among other project tasks [6]. To accomplish this, a vital aspect of using the Kanban process is visualizing the work being done and WIP bottlenecks [19] in the form of a Kanban board [19], [20] with the intention of quickly producing a value-added product/service for the customer. Some of the Kanban method's essential characteristics include the lack of any prescribed roles, continuous delivery, work being "pulled" through the system, changes that can be made at any time, lead time that is used as a default metric for planning and process improvement, commitment optional, WIP limits directly (per workflow state), and a Kanban board is persistent and shared among multiple teams [2], [12], [13], [21].

Two major disadvantages of high WIP levels that are difficult to economically evaluate are not being able to respond to demand changes quickly and the potential to build a considerable quantity of poor-quality stock before realizing that there is a quality problem. To help control inventory within production and manufacturing facilities, WIP limiting production procedures are frequently used [22].

Limiting the amount of WIP is considered a fundamental principle in Kanban software development [23]. There are several ways to set WIP limits, including Brechner's approach [24] or the floor–ceiling [25] approach.

Kanban uses a mechanism known as a pull system, as new work is "pulled" into the system when capacity exists to handle it, rather than being pushed into the system based on demand. A pull system cannot be overloaded if the capacity, as fixed by the number of signal cards in circulation, has been set appropriately [15].

III. Kanban Method

Today, one can find the Kanban method being implemented all over the world in various organizations with the aim of achieving continuous improvements in their process management. In addition to Kanban core practices like visualize workflow, set WIP limits, and manage flow, Anderson [15] added three more for

Backlog	Ready (4)	Analyse (2)		Develop (3)		Test (2)	
		Doing	Done	Doing	Done	Doing	Done

Fig. 1. Kanban board used.

software development, i.e., make policies explicit, implement feedback loops, and drive improvement based on methods and models. In the following sections, these practices are briefly introduced.

A. Visualize Workflow

Visualize workflow is a fundamental element of Kanban and entails creating a board that shows the discussed process' stages. Each stage indicates the current state of a work item. The Kanban board used for software development is typically divided into stages or phases, such as ″Backlog," ″Analyze," ″Develop," ″Test″ and ″Deploy." Each stage occupies one column on the board, further divided into two subcolumns: 1) ″Doing″ and 2) ″Done." A work item is represented by a card or sticky note placed on a certain subcolumn.

Most Kanban teams rely on the T-shirt technique to estimate work items' sizes in S, M, L (small, medium, large) and try to partition the items into similar small sizes. Work items also come in different types, such as standard, expedite, fixed date, and intangible. Each type typically has its own color.

In our example, we used standard work items because these represent the vast majority on the board. Fixed-date items are planned in advance and sorted in the backlog in order to be pulled into the board at the right time to meet their deadline. There are typically a small number of expedite items since there may only be one expedite item on the board at a time. Intangible items should also be low in number or else the team may need to deal with the issue of quality on the board. In addition, we relied on data from Brechner's example [24], which uses only standard ones.

Boards are visual control systems that allow teams to visually observe the WIP and to self-organize, assign their own tasks, and move work from "backlog to completed" without direction from a project or line manager [15].

Fig. 1 shows our board, structured as follows: Backlog, Ready, Analyze, Develop and Test. The board is simulated, making it difficult to indicate the flow of work items across the board on a certain day.

B. Limit WIP

Limiting WIP is an essential principle that makes the pulling system work. This means every team member must finish her/his current work before pulling another work item into the next stage. If there is no explicit limit to WIP and no signaling to pull new work through the system, the system is not Kanban [15].

A key aspect of Kanban is to reduce the amount of multitasking that many teams and knowledge workers are prone to do. The WIP limit defined at each stage of the board encourages team members to finish the work at hand and only then to take up the next piece of work.

Controlling the flow of development using WIP limits promotes continuous development without wasting resources. The Kanban visualizes the overall flow of development and the state of the tasks and continues to improve the process [26]. The process flow is a dynamic process that starts when an input (work item) enters a process, continues processing throughout different stages of the process, and ends when it leaves the process as its output [27].

There is causation between the quantity of WIP and the average lead time, and the relationship is linear. In the manufacturing industry, this relationship is known as Little's law [15]. The law states that the average WIP or inventory (I) equals the average flow rate (R) times the average lead time (T), as follows:

$$I = R \times T.$$

Little's law expresses that the more work items are put into the process, the longer the lead time of each item will take to pass through. Therefore, to create an effective process with the shortest lead time, we must define the correct WIP limit for each stage in the process. Teams usually define WIP limits at the start of a project and, then, change them as the work proceeds in order to improve the process and reduce the possibility of queues being created.

C. Manage Flow

The main reported benefits of using the Kanban method were improved lead time to deliver software, improved quality of software, improved communication and coordination, increased consistency of delivery, and decreased customer reported defects [28]. The above-mentioned benefits depend on a continuous flow state on the board. Therefore, the role of the manage flow practice is to prepare an environment for achieving the continuous flow concept; this is a process flow without delays or waiting in queues. This process flow is created by removing wastes and nonvalue-added activities, making the team focused on value-added work and, thus, developing a value-added product for the customer.

Toyota identified seven categories of waste in the manufacturing process, whereas Poppendieck and Poppendieck [29] determined seven categories of waste in software development, which are partially done work, extra features, relearning, handoffs, delays, task switching, and defects.

The Kanban team manages the flow of work items via different activities like the "definition of done″ and different meetings, such as stand-up, replenishment, and deployment. A stand-up is a 15-min meeting organized every morning in front of the board and is the only meeting required by Kanban. In addition, Kanban teams attend replenishment and deployment meetings. The purpose of a replenishment meeting is to determine how many items are entering the process each week. The deployment meeting is important to decide for deploying completed items to the customer. These activities permit the team to accelerate the flow, ensure faster delivery, and make the project predictable.

D. Make Process Policies Explicit

Policies are clear rules defined in the board's stages to constrain the flow of work items. The policy should be visible and defined in each stage, understandable, and changeable whenever needed. Examples of policies include Acceptance Criteria, Definition of Done, Replenishment, and Delivery Cadence.

E. Implement Feedback Loops

Organizing feedback loops and getting information from the customer about the delivered features is required to implement changes and improvements in the team's work. Feedback also triggers discussions and collaboration among team members and also with the customers. Organizing an effective delivery cadence and reviews is essential for obtaining the customer's feedback and keeping them involved in the work of the Kanban team.

F. Drive Improvement Based on Methods and Models

In the Kanban method, we start with the organization as it is in reality and encourage teams to make small, continuous, and incremental improvements. This helps develop an effective process and, thus, creates the possibility of a continuous flow. For this purpose, Kanban is open to the use of any technique, method, or model for process optimization.

IV. Simulation Approach

The article applied a simulation approach as a research method and ran simulation scenarios using the Arena Simulation software. This approach was pursued in order to respond to the ever-challenging task of setting suitable WIP limits due to heavy influence on creating a sustainable workflow across the board. The primary motivation for running the simulation scenarios in this way was to ensure to not set WIP limits that are too low, as that could result in people being idle, nor too high since that may result in idle work.

The Kanban method relies on the pulling system concept, which means that any completed work item that is momentarily in the column of a certain stage is pulled into the next stage by an available resource (team member) if the WIP limit of the next stage is greater than the number of the work items already inside in its column.

To introduce the approach, the result of the example found in Brechner [24] was used. This result suggested a team of one analyst, three developers, and two testers; and WIP limit values of 2, 5, and 3 defined in the sequential stages Analyze, Develop, and Test.

The proposed approach has the following three steps: 1) generate all combinations, 2) determine the replenishment value, and 3) run board simulations.

A. The Approach

The two approaches mentioned in Section II, discuss the problem of WIP limit setting. However, the aim of this article is to introduce a new approach for not only dealing with setting WIP limits but also with finding the best relationship solution among three parameters: 1) replenishment value, 2) WIP limit, and 3) resource capacity. Comparing this approach with other simulation approaches exceeds the scope of this article.

Step 1: Generate all combinations. This step starts by identifying two groups of values, where the first group contains numbers of resources (team members) assigned to the board stages and the second group consists of WIP limit values suggested by the team. The step continues by generating all possible WIP limit combinations, starting with a resource numbers combination and ending with a WIP limit values combination. The combinations number in our example is 12, which starts with the first WIP limits combination that is, 1, 3, 2 (number of resources in different stages), and the last one, thus, is 2, 5, 3 (number of WIP limit values). Table I shows all the WIP limit combinations generated.

TABLE I
WIP Limit Combinations for Simulation Run 1

WIP limit Combination	Analyse	Develop	Test
1	1	3	2
2	1	3	3
3	1	4	2
4	1	4	3
5	1	5	2
6	1	5	3
7	2	3	2
8	2	3	3
9	2	4	2
10	2	4	3
11	2	5	2
12	2	5	3

Step 2: Determine the replenishment value. Replenishment involves determining the number of work items to be entered in Ready per period of time, usually a week. The team agrees on replenishment value(s) at a replenishment meeting arranged for this purpose. These values are later tested using a simulation technique to establish the most suitable one, namely, the replenishment, which minimizes the queues on the board and has the shortest lead time.

In our example, we decided to test two replenishment values: 2 items per week and 3 items per week. The replenishment value should be reasonably higher than the resources' number assigned in the first stage of the board, in our case the value is greater by 1 or 2. Any bigger value may create work idle and create a queue in Ready. From the other side, if the selected replenishment value per week equals the resource number in the first stage, people's idleness might follow.

Step 3: Run board simulations. Simulation is a technique that enables the modeling and launching of an imitation of a certain process' functionality so as to identify its performance without disturbing work in the original process. A critical property of the simulation is that it enables experts to conduct experiments on the system's behavior by using "what-if" questions.

Fujimoto [30] defined simulation as "a system that represents or emulates the behavior of another system over time." Banks and Gibson [31] referred to simulation as "the imitation of

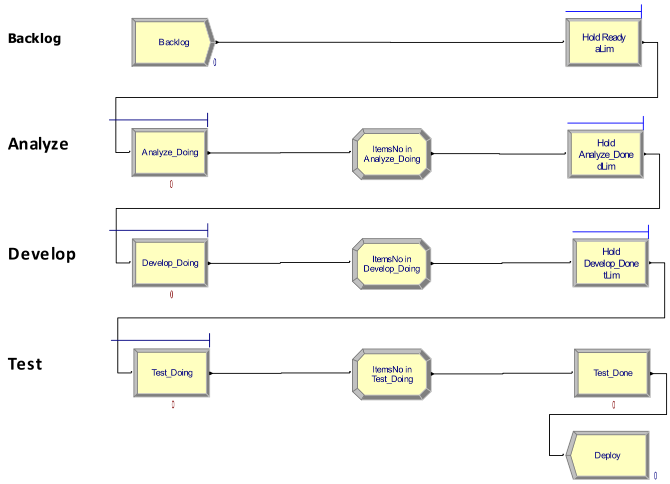

Fig. 2. Simulation model.

the operation of a real-world process or system over time." Furthermore, Laguna and Marklund [32] defined simulation simply as a "means to mimic the reality in some way."

In this step, for each replenishment value determined in the previous step, a board simulation is run as many times as the number of WIP limit combinations generated in the first step.

In our case, we executed the board simulation 24 times for the following:

1) a replenishment of 2 items/week, the simulation of the board was run 12 times for the WIP limit combinations in Table I;
2) a replenishment of 3 items/week, the simulation of the board was run 12 times again for the WIP limit combinations presented in Table I one-by-one.

The board simulation was run by considering the following assumptions:

1) the replication length is 300 days;
2) hours per day are 8;
3) the replenishment values used are 2 items/week and 3 items/week;
4) the simulation every time uses a new WIP limit combination from the generated ones;
5) the team consists of one analyst, three developers, and two testers;
6) the duration of the Analyze stage is 2–3 days;
7) the duration of the Develop stage is 3–5 days;
8) the duration of the Test stage is 2–3 days;
9) an available resource can pull a work item into the next stage only if the number of work items in the next stage is less than the WIP limit set for this stage.

Fig. 2 shows a model comprising a few modules (see [33]), which are as follows.

1) The create module generates arrival items in the model; in our case, it is labeled "Backlog."
2) The dispose module is used to end the simulation started; in our model, it is named "Deploy."
3) The hold module retains items until a certain condition is satisfied. Two conditions apply to hold modules: First, the WIP limit defined in the next stage be greater than the number of work items in the stage, and second, a team member is available to pull the new item into the stage. For example, "Hold Ready aLim″ holds items until, first, the WIP limit (aLim) in the "Analyze″ stage is greater than the number of items in the stage, and second, an analyst is available.
4) The process module enables the resource to execute the work defined in it. An analyst, for instance, analyzes the item pulled into the "Analyze_Doing″ process.
5) The assign module is used to assign a value to an attribute. Our module "ItemsNo in Analyze″ calculates the number of items in the stage "Analyze″ ("Analyse_Doing″ and "Hold Analyze_Done dLim″). If both conditions defined in the ″Hold Read aLim″ module are satisfied, then it allows an item to leave.

TABLE II
Results for Simulation Run 1

Columns \ Parameters	Simulation	Replenishment 2 Items/Week	Replenishment 3 Items/Week
Board	In	86	129
	Out	84	117
	WIP	3	9
	Work Time	9.1	9.1
	Wait Time	1.2	11.1
	Lead Time	10.3	20.3
Ready	Ready (wait time)	1.2	18.99
Analyse	Doing (time)	2.5	2.5
	Done (wait time)	0	0
Develop	Doing (time)	4.1	4.2
	Done (wait time)	0	0
Test	Doing (time)	2.5	2.5
	Done (wait time)	0	0
Utilisation	Analyst	71%	99.8%
	Developer 1	40%	56%
	Developer 2	38%	55%
	Developer 3	39%	54%
	Tester 1	34%	48%
	Tester 2	36%	49%

B. Simulation Run 1

First, the board simulation was run 12 times using the WIP limit combinations listed in Table I one-by-one and replenishment of 2 items/week. Second, the simulation of the board was again executed 12 times using a replenishment of 3 items/week and the same WIP limit combinations as the first time. The results of these simulations are shown in Table II. It shows that we gained only one solution for each of the two replenishment values, presented in the third and fourth columns of the table, despite changing the WIP limit combination for each simulation run.

Analysis of the first solution reveals the following characteristics.

1) It creates only 86 work items that enter the board due to the low replenishment value, of which 84 work items exited the board.
2) The average number of WIP across the board is only three work items, which is very small.

3) The average lead time for a work item to pass through the board is 10.3 days, which is a good result.
4) Resource utilization is analyst 71%, developers $\leq$ 40%, and testers $\leq$ 36%, which means the developers and testers are engaged in some idleness.

These characteristics make it clear that, despite the short lead time, the solution is not useful because the replenishment of two items/week creates a small number of work items (inputs), which leads to people being idle.

Analysis of the second solution indicates the following characteristics.

1) A replenishment of three items/week created 129 work items that entered the board, of which 117 work items left the board.
1) The average number of WIP within the board is nine work items, which is a big number and means more work items are waiting in queues.
2) The average lead time is 20.3 days, which is too long a time and caused by the long average waiting time on the board of 11.1 days.
3) Resource utilization is analyst 99.8%, developers $\leq$ 56%, and testers $\leq$ 49, which means the analyst is overloaded, while the developers and testers still engage in some idleness.

This solution is, therefore, also not useful since the analyst is overloaded, consequently leading to queues and a long lead time.

Analysis of the first solution described above shows that the reason for obtaining this less useful solution is the replenishment of 2 items/week, which meant to few work items (inputs). Meanwhile, analysis of the second solution showed that using only one analyst creates a bottleneck that makes the board inefficient due to the analyst being unable to handle the analysis of so many inputs.

C. Simulation Run 2

We found the only way to improve the process efficiency is to change the team structure and adjust it so that analysts have more work items. To solve this bottleneck and make work items pass through the board more smoothly, a new analyst is added to the team, hence the new team consists of two analysts, three developers, and two testers.

Moreover, the replenishment and WIP limit values must be adjusted to the changes made to the number of analysts. Accordingly, the new replenishment values to be used are three items/week, four items/week, and five items/week.

Our work continued by generating all combinations starting with the combination of the changed resource numbers (2, 3, 2) and ending up with updated WIP limit values (3, 5, 3). The WIP limit in the Analyze stage was changed to 3, namely, analysts number + 1. Table III shows the new WIP limit combinations.

The simulation procedure is repeated 24 times using the WIP limit combinations listed in Table III; thus, the board simulation was run 12 times for each replenishment value. The results of these simulations are shown in Table IV.

Table IV presents the three new solutions that emerge using the new simulation environment created by relying on the changed replenishment values and new WIP limit combinations shown in Table III.

TABLE III
WIP Limit Combinations for Simulation Run 2

WIP limit Combination	Analyse	Develop	Test
1	2	3	2
2	2	3	3
3	2	4	2
4	2	4	3
5	2	5	2
6	2	5	3
7	3	3	2
8	3	3	3
9	3	4	2
10	3	4	3
11	3	5	2
12	3	5	3

TABLE IV
Results for Simulation Run 2

	Parameters	Replenishment 3 Items/Week	Replenishment 4 Items/Week	Replenishment 5 Items/Week
Board	In	129	172	215
	Out	126	167	208
	WIP	4	6	8.6
	Work Time	9	9	9
	Wait Time	0.9	1.6	3
	Lead Time	9.9	10.6	12.1
Ready	Ready (wait time)	0.8	1.3	2.1
Analyse	Doing (time)	2.5	2.5	2.5
	Done (wait time)	0	0.3	0.6
Develop	Doing (time)	4	4	4
	Done (wait time)	0.09	0.08	0.4
Test	Doing (time)	2.5	2.5	2.5
	Done (wait time)	0	0	0
Utilisation	Analyst 1	55%	72%	91%
	Analyst 2	54%	71%	87%
	Developer 1	57%	76%	94%
	Developer 2	58%	77%	94%
	Developer 3	55%	76%	94%
	Tester 1	55%	71%	87%
	Tester 2	51%	68%	88%

A comparison of these three solutions shows that the second solution is superior and the board workflow became smoother due to the following:

1) average WIP number of six, which is suitable given the number of resources, while it is too low in the first and too high in the third solution;
2) average resource utilization shows the team is sufficiently busy and any rise in utilization may lead to team burnout. Otherwise, it is low in the first solution, which revealed people were idle, and unrealistically high in the third solution, which led to work idleness (longest average waiting time).

The lead time of 10.6 days is 0.7 of a day longer than that in the first solution, yet this is meaningless when compared to the first two parameters (average WIP and resource utilization) and is 1.5 days shorter than that in the third solution.

V. Discussion

The Kanban method enables teams to achieve a continuous workflow, namely, one without (or minimal) queues or people being idle in different stages of the board, thereby assuring the uninterrupted delivery of value-added items to the customer.

To achieve a workflow of this nature, important steps of the Kanban method must be taken, such as creating a well-visualized board, defining a suitable replenishment value, setting proper WIP limits, and assigning the resource capacity needed in various board stages. Such steps are in addition to the small changes, improvements, and fresh ideas the team makes continuously during the process flow.

The literature review pointed to the important issue of whether, simply by adjusting the WIP limits, teams can create sustainable workflows with minimal waiting time and a reasonable number of work items on the board without creating work or people's idleness. Based on the run simulations presented in Section IV, we believe that the issue is complex since merely adjusting the WIP limits proves to be insufficient without connecting it to the other parameters that affect the flow performance. We show that generating an effective workflow depends on finding the optimal relationship, which entails the right balance of parameters like, replenishment value, WIP limits, and resource capacity, which in turn will create a stable system on the board.

The Kanban team reduces risk on the board by creating small-sized work items in order to complete them within a similar time frame, minimize waste, and focus on value-added items while making minor changes whenever needed.

Kanban encourages gradual process improvements by organizing retrospective meetings. Although Kanban does not require a retrospective meeting, many teams arrange one every week or as needed to discuss, evaluate, and improve their ways of working. Analyzing the results of the three solutions presented in Table IV shows that using the simulation approach helps Kanban teams find the optimal workflow on the board based on characteristics of the best solution. The second solution that is obtained by using a replenishment value of four items/week proved that it is the best of all three tested solutions because it generated a WIP of six items, which is in line with the number of team members used and the WIP limits defined in the stages. Consequently, the lead time (10.6 days) ensured the completion of 167 work items, and a sufficiently busy team (utilization $>$ 70%), makes this solution optimal, realistic, and acceptable.

The pull system is a flow-controlling technique that allows an available team member to pull only one work item into the next stage, thereby allowing the team to focus on ensuring quality work and achieving a continuous flow. The simulation model shown in Fig. 2 successfully imitates Kanban's pulling principle, starting from the last stage on the board and moving stage by stage toward the first one. In each stage, two conditions are tested to pull an item into the next stage: 1) resource availability and 2) the WIP limits are greater than the WIP in the stage.

Unlike the discussed way of WIP limits setting in these two approaches, our approach generates a sustainable workflow on the board by determining the optimal relationship between different values of replenishment, resource capacity, and WIP limits.

As mentioned above, approaches exist like Brechner's [24] or the floor–ceiling approach [25]. The former approach is based on computing the WIP limits by calculating the throughput in the slowest step on the board and adjusting the throughput in the other steps to that. However, the second approach is based on the use of two strategies, that is, the floor strategy whereby the team starts with small limit values and raises them as required, and the ceiling strategy where the team starts with higher values and decreases them as required. Unlike the discussed way of WIP limits setting in these two approaches, our approach generates a sustainable workflow on the board by determining the optimal relationship between different values of replenishment, resource capacity, and WIP limits.

VI. Conclusion

The purpose of this article was to introduce a solution to overcome the problem of efficiently managing the Kanban board workflow by minimizing queues in the stages, which was achieved by allocating the resource capacity with work items and WIP limits.

By employing such a research method, the presented solution contributed in several ways to the Kanban method theory and practice. First, it proposed a new efficient approach to the theory and the overall debate on Kanban by triggering new research that addressed the typical challenges Kanban teams face in generating a continuous flow. Second, the approach proved that generating a sustainable workflow on the Kanban board is complex and depends on finding the optimal relationship among replenishment, WIP limits, and resource capacity. This approach differed in concept from already existing approaches that solely focus on setting better WIP limits. Third, the results indicated that this approach can be used in various fields, such as process management, supply chain management, IT project management, and enterprise agile transformation, namely, wherever Kanban is implemented. The fourth important contribution was the development of a simulation model that imitates the Kanban board and the pull system, starting from the last stage moving toward the first. Finally, our approach created an effective Kanban board that may influence particularly ineffective management, one that is focused on maximizing the utilization of resources, to change their behavior, and to accept and begin to apply agile management principles and methods.

The presented approach solved this problem by using a simulation technique. In our example, we ran the board simulation 60 times, in the first 24 simulations using one analyst, two replenishment values (two items/week and three items/week), and the WIP limit combinations shown in Table I. In the second, 36 simulations were run using two analysts, three replenishment values (three items/week, four items/week, and five items/week), and a new group of WIP limit combinations listed in Table III.

The simulations results allowed the following conclusions to be drawn. First, changing only the WIP limits cannot lead to a breakthrough in terms of creating an effective workflow with/without minimal queues and people's idleness because ultimately an available resource always needs to be there in order to pull the work item into the stage. Second, to achieve an important improvement in the workflow and shorten the lead time, it is essential to find a harmonizing, optimal relationship solution among the three parameters of replenishment value, resource numbers in stages, and a WIP limit combination. Third, such a harmonious solution can be found by running the simulation using different possible suggested replenishment values, resource capacities, and WIP limit combinations. Fourth, the introduced approach solves the difficult problem of finding a suitable WIP limit combination and makes it an easy task. Therefore, it is hoped that the approach will help spread the use of the method in practice by many agile teams.

The proposed approach is limited to implementing the pull system, which means it is suitable for Kanban and could be used with similar kinds of methods. Moreover, we believe that the developed approach will trigger new research dealing with Kanban workflow and could initiate the development of similar approaches for other agile methods.

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